Adversarial Fairness

Qualitative & Analytic Report

Dataset: earnings_synth

2026-06-26

Accuracy: 89.3% -> 86.3%
Iterations: 5
origin_group: PASS — P-rule 86% ≥ 80%
background_group: PASS — P-rule 83% ≥ 80%

Qualitative analysis generated by LLM: llama3.2

Qualitative analysis (LLM)

The classifier's predictions show that the disadvantaged group for 'origin_group' is A, with a predicted-positive rate of 22.1% and an actual outcome rate of 25.5%, resulting in a gap of 3.4 percentage points (P-rule) or approximately 0.69 times the original gap. In contrast, the disadvantaged group for 'background_group' is Y, with a predicted-positive rate of 19.0% and an actual outcome rate of 13.6%, leading to a gap of 5.4 percentage points (P-rule) or about 1.14 times the original gap. The model's debiasing efforts have reduced these gaps, but at the cost of accuracy, with the P-rule improving from 85.8% to 82.5%. However, the actual outcome-rate gap remains unchanged after debiasing, indicating that the model does not amplify or reduce this disparity; it merely adjusts its predictions to minimize the difference between predicted and actual outcomes. The gaps in these sensitive attributes are not designed biases but rather reflect correlations with other proxies that the model has learned from the data.

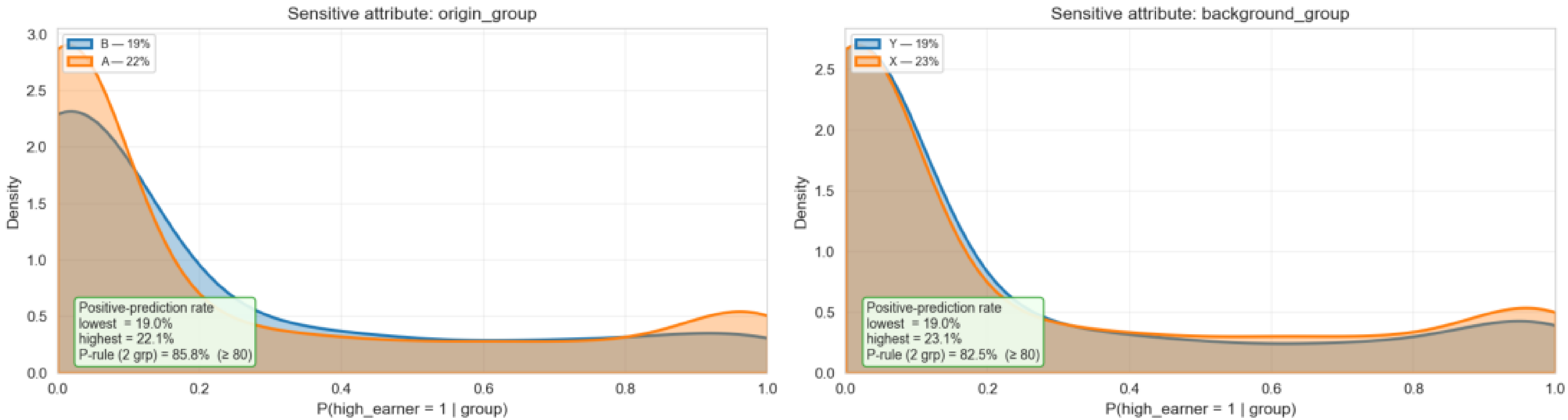
Fairness metrics (before -> after)

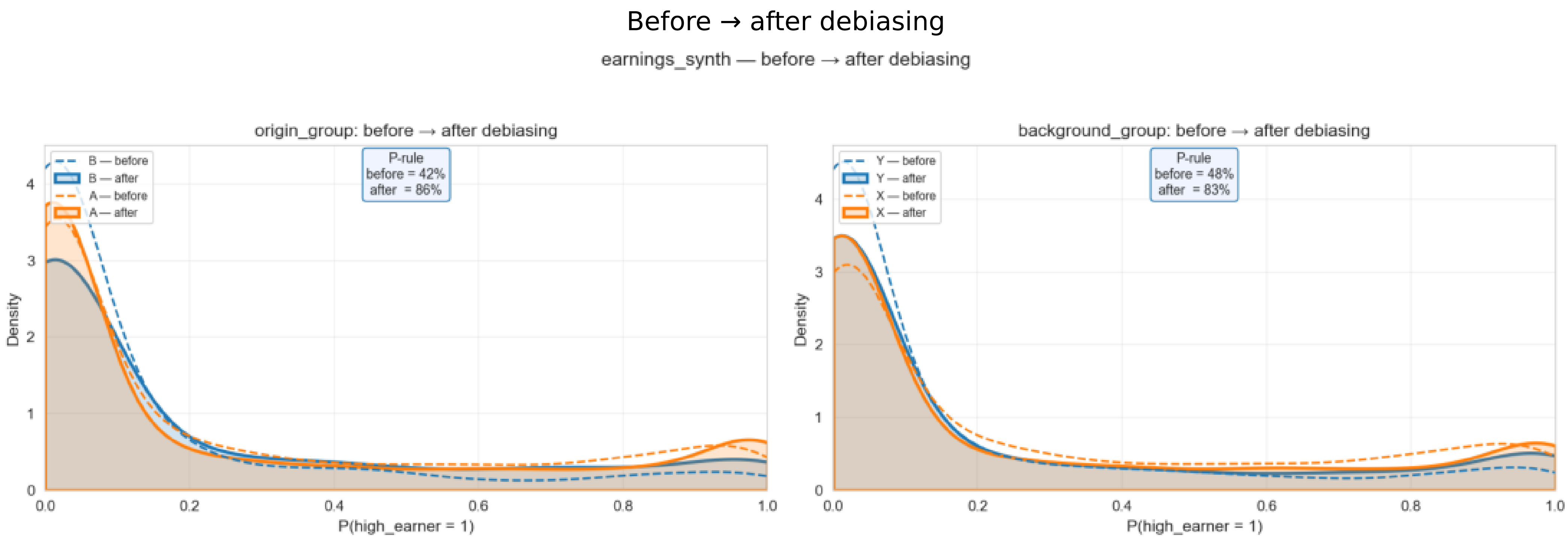
attribute	P-rule	DP	EoDD	EoOpp
origin_group	42.3 -> 85.8	13.4 -> 3.1	4.6 -> 10.1	6.5 -> 15.2
background_group	47.5 -> 82.5	13.6 -> 4.0	5.5 -> 7.3	8.2 -> 11.0

P-rule higher is fairer; DP / EoDD / EoOpp lower is fairer.

Predicted-outcome distribution by group

earnings_synth — predicted P(high_earner = 1) by sensitive group




Before → after debiasing
earnings_synth — before → after debiasing
origin_group: before → after debiasing
B — before
B — after
A — before
A — after
P-rule
before = 42%
after = 86%
Density
P(high_earner = 1)
background_group: before → after debiasing
Y — before
Y — after
X — before
X — after
P-rule
before = 48%
after = 83%
Density
P(high_earner = 1)

Equalized odds — before vs after